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Functional Annotation with SnpEff & bcftools

Objective

To perform downstream functional annotation of pre-called genetic variants using SnpEff, benchmark alternative variant discovery via VarScan from read pileups, and detect mutations/structural rearrangements in microbial datasets utilizing breseq

1. SnpEff Pipeline

Principle
SnpEff is a variant annotation tool. It takes a VCF containing identified variants and compares their genomic positions with a genome annotation database to determine the functional effect of each variant, such as synonymous, missense, nonsense, frameshift, etc.
Starting Input
Input: VCF file generated from the previous GATK variant-calling workflow.

Workflow:
Final VCF
↓
Check SnpEff database
↓
Check genome/reference compatibility
↓
Rename chromosome IDs if required
↓
SnpEff annotation
↓
Annotated VCF
↓
Extract ANN information
↓
Identify HIGH / MODERATE / LOW effects
↓
Functional interpretation

1. SnpEff Annotation & Variant Profiling

Step 1 — Inspect the Input VCF

Before annotating, inspect the uncompressed lines and header fields of the variant call file:

Bash

```
bcftools view -H conA_variants_final.vcf | head -3
```

```
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$ snpEff -stats conA_snpEff_summary.html   R64-1-1.99   conA_variants_snpEff_input.vcf > conA_variants_annotated.vcf
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$ bcftools view -H conA_variants_final.vcf | head -3
NC_001133.9     32145   .       C       T       71.32   .       AC=2;AF=1;AN=2;DP=2;ExcessHet=0;FS=0;MLEAC=1;MLEAF=0.5;MQ=60;QD=25.36;SOR=0.693 GT:AD:DP:GQ:PL   1/1:0,2:2:6:83,6,0
NC_001133.9     32511   .       C       T       110.84  .       AC=2;AF=1;AN=2;DP=3;ExcessHet=0;FS=0;MLEAC=1;MLEAF=0.5;MQ=60;QD=28.73;SOR=1.179 GT:AD:DP:GQ:PL   1/1:0,3:3:9:124,9,0
NC_001133.9     33548   .       A       C       1064.06 .       AC=2;AF=1;AN=2;DP=27;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=30.97;SOR=6.666  GT:AD:DP:GQ:PL   1/1:0,27:27:81:1078,81
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$
```

bcftools view -h conA_variants_final.vcf | grep "^##reference"
bcftools view -h conA_variants_final.vcf | grep "^##contig"
bcftools view -h conA_variants_final.vcf | grep "^##source"

```
bcftools view -h conA_variants_final.vcf | grep "^##contig"
bcftools view -h conA_variants_final.vcf | grep "^##source"
##contig=<ID=NC_001133.9,length=230218>
##contig=<ID=NC_001134.8,length=813184>
##contig=<ID=NC_001135.5,length=316620>
##contig=<ID=NC_001136.10,length=1531933>
##contig=<ID=NC_001137.3,length=576874>
##contig=<ID=NC_001138.5,length=270161>
##contig=<ID=NC_001139.9,length=1090940>
##contig=<ID=NC_001140.6,length=562643>
##contig=<ID=NC_001141.2,length=439888>
##contig=<ID=NC_001142.9,length=745751>
##contig=<ID=NC_001143.9,length=666816>
##contig=<ID=NC_001144.5,length=1078177>
##contig=<ID=NC_001145.3,length=924431>
##contig=<ID=NC_001146.8,length=784333>
##contig=<ID=NC_001147.6,length=1091291>
##contig=<ID=NC_001148.4,length=948066>
##contig=<ID=NC_001224.1,length=85779>
##source=HaplotypeCaller
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$
```

- The -H flag strips out metadata lines to reveal raw variant calls.
- The -h flag isolates the header, allowing verification of contig labels and caller provenance (e.g., GATK HaplotypeCaller).

Step 2 — Verify and Install SnpEff

Ensure SnpEff is set up in your working environment:

Bash

conda install -c bioconda -c conda-forge snpeff=5.2
snpEff -version

Step 3 & 4 — Search and Fetch the Reference Database

Confirm whether the target organism database is present locally or in the remote repository, then download it:

Bash

wget https://snpeff-public.s3.amazonaws.com/databases/v5_0/snpEff_v5_0_R64-1-1.99.zip
unzip snpEff_v5_0_R64-1-1.99.zip

- For yeast (*Saccharomyces cerevisiae*), the pre-built reference is R64-1-1.99.

Step 5 — Inspect Database Contig Naming

Check how chromosomes are formatted inside the SnpEff database:

Bash

snpEff dump -dataDir "$PWD/data" R64-1-1.99 | head -n 50



- The database employs Roman numerals (e.g., I, II, IV) rather than NCBI RefSeq accessions (NC_001133.9).

  Step 6 & 7 — Harmonize Chromosome Identifiers

  Create a two-column, tab-separated translation table (chr_map.txt) and rewrite the VCF headers:

  Bash

  nano chr_map.txt
  (Map old IDs to new IDs, for instance: NC_001133.9 I). As our input VCF file names the yeast chromosomes using NCBI accession numbers like this:

  Plaintext

  NC_001133.9
  NC_001134.8
  NC_001135.5
  However, the **SnpEff database (R64-1-1.99)** only recognizes Roman numerals:

  Plaintext

  I
  II
  II

  So we can enter directly via this command:

  cat << 'EOF' > chr_map.txt

  NC_001133.9 I

  NC_001134.8 II

  NC_001135.5 III

NC_001136.10 IV

NC_001137.3 V

NC_001138.5 VI

NC_001139.9 VII

NC_001140.6 VIII

NC_001141.2 IX

NC_001142.9 X

NC_001143.9 XI

NC_001144.5 XII

NC_001145.3 XIII

NC_001146.8 XIV

NC_001147.6 XV

NC_001148.4 XVI

NC_001224.1 Mito

EOF

Apply the translation via bcftools annotate:

Bash

bcftools annotate --rename-chrs chr_map.txt -O v -o conA_renamed.vcf conA_variants_final.vcf

Step 8 — Verify the Renamed VCF

Confirm that the chromosome entries were successfully transformed:

Bash

bcftools view -H conA_renamed.vcf | head -5

```
NC_001148.4     XVI
NC_001224.1     Mito
EOF
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$ bcftools annotate --rename-chrs chr_map.txt -O v -o conA_renamed.vcf conA_variants_final.vcf
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$ bcftools view -H conA_renamed.vcf | head -5
I       32145   .       C       T       71.32   .       AC=2;AF=1;AN=2;DP=2;ExcessHet=0;FS=0;MLEAC=1;MLEAF=0.5;MQ=60;QD=25.36;SOR=0.693 GT:AD:DP:GQ:PL   1/1:0,2:2:6:83,6,0
I       32511   .       C       T       110.84  .       AC=2;AF=1;AN=2;DP=3;ExcessHet=0;FS=0;MLEAC=1;MLEAF=0.5;MQ=60;QD=28.73;SOR=1.179 GT:AD:DP:GQ:PL   1/1:0,3:3:9:124,9,0
I       33548   .       A       C       1064.06 .       AC=2;AF=1;AN=2;DP=27;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=30.97;SOR=6.666  GT:AD:DP:GQ:PL   1/1:0,27:27:81:1078,81,0
I       33612   .       A       G       572.06  .       AC=2;AF=1;AN=2;DP=15;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=27.24;SOR=5.421  GT:AD:DP:GQ:PL   1/1:0,14:14:42:586,42,0
I       33700   .       G       A       321.01  .       AC=2;AF=1;AN=2;BaseQRankSum=1.645;DP=12;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;MQRankSum=0;QD=32.1;ReadPosRankSum=1.645;SOR=3.258
T:AD:DP:GQ:PL   1/1:1,9:10:20:335,20,0
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

Step 9 — Run Functional Annotation

Execute SnpEff using the mapped VCF against the database:

Bash

snpEff -dataDir "$PWD/data" R64-1-1.99 conA_renamed.vcf > conA_annotated.vcf

Output:
I 32145 C T 71.32 T|synonymous_variant|LOW|GDH3|YAL062W|transcript|
YAL062W_mRNA|protein_coding|1/1|c.579C>T|p.Ala193Ala|579/1374|579/1374|193/457||,T|
upstream_gene_variant|MODIFIER|FLO9|YAL063C|transcript|YAL063C_mRNA|protein_coding||

c.-4177G>A|||||4177|,T|upstream_gene_variant|MODIFIER|BDH2|YAL061W|transcript|YAL061W_mRNA|protein_coding||c.-1303C>T|||||1303|,T|upstream_gene_variant|MODIFIER|BDH1|YAL060W|transcript|YAL060W_mRNA|protein_coding||c.-3010C>T|||||3010|,T|upstream_gene_variant|MODIFIER|ECM1|YAL059W|transcript|YAL059W_mRNA|protein_coding||c.-4364C>T|||||4364|,T|downstream_gene_variant|MODIFIER|YAL059C-A|YAL059C-A|transcript|YAL059C-A_mRNA|protein_coding||c.*4351G>A|||||4351|

Detailed Variant Breakdown (Position Chr I: 32145)

- **Genomic Coordinate:** Chromosome I, 32145
- **Reference** / **Alternate Allele:** C -> T
- **Quality Score (QUAL):** 159.14 (high-confidence call)
- **Gene & Transcript:** *GDH3* (YAL062W)

- **Coding & Protein Change:** c.579C>T, resulting in p.Ala193Ala
- **Predicted Consequence: synonymous_variant** (LOW impact). Because alanine is retained in the translated polypeptide, this mutation is functionally neutral and does not disrupt enzymatic performance.

Step 10 & 11 — Confirm Functional Annotations

Verify that SnpEff successfully injected ANN subfields into the INFO string:

Bash

```
bcftools view -h conA_annotated.vcf | grep "ID=ANN"
bcftools view -H conA_annotated.vcf | head -5
```



Step 12 — Check for Unmatched Chromosomes

Ensure zero discrepancies between the VCF contigs and the database build:

Bash

```
bcftools view -H conA_annotated.vcf | grep -c "ERROR_CHROMOSOME_NOT_FOUND"
```

- **Expected Output:** 0 (any positive integer denotes unmapped records that were skipped during annotation).

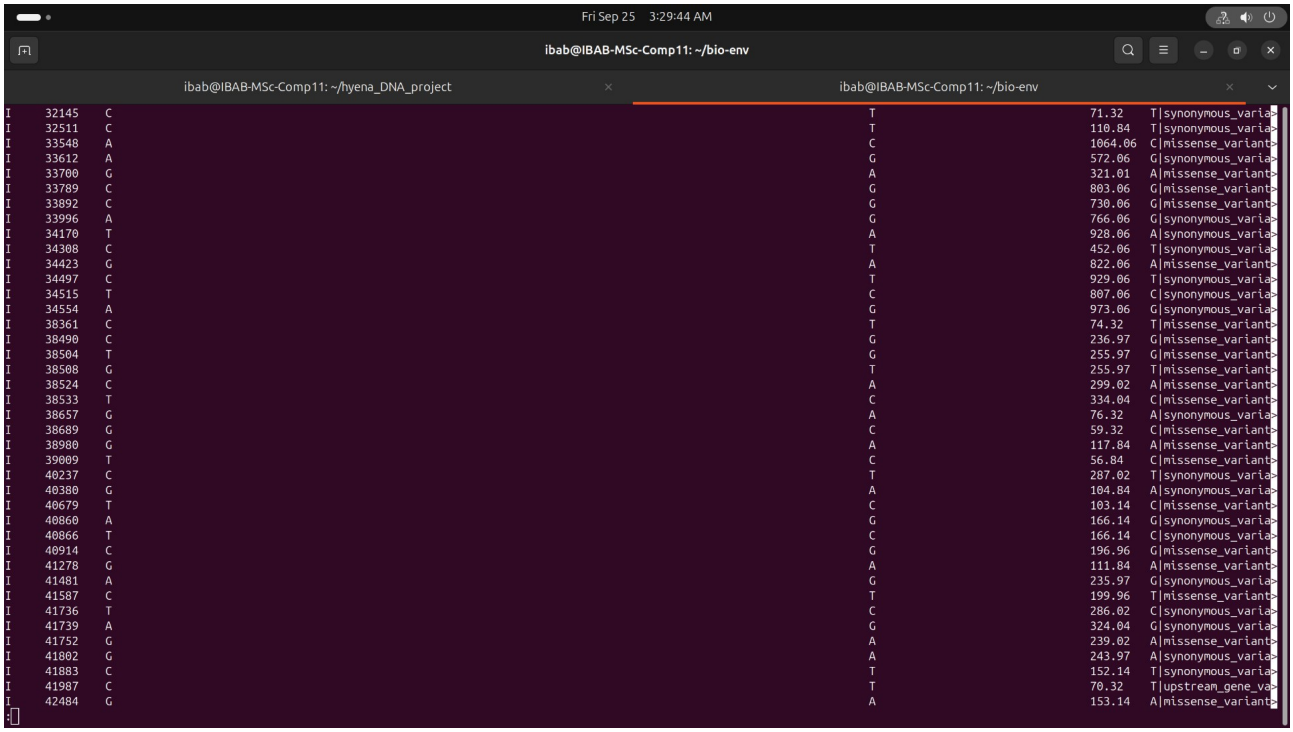
```
m_gene_variant|MODIFIER|YAL059C-A|YAL059C-A|transcript|YAL059C-A_mRNA|protein_coding||c.*2796C>T|||||2796|,A|downstream_gene_variant|MODIFIER|YAL056C-A|YAL056C-A|transcript|Y
otein_coding||c.*4996C>T|||||4996|      GT:AD:DP:GQ:PL  1/1:1,9:10:20:335,20,0
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$ bcftools view -H conA_annotated.vcf | grep -c "ERROR_CHROMOSOME_NOT_FOUND"
0
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

Step 13 & 14 — Extract and Inspect Formatted Tables

Parse the annotated VCF into a readable tabular summary:

Bash

```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%QUAL\t%INFO/ANN\n' conA_annotated.vcf > conA_annotation.txt
column -t conA_annotation.txt | less -S
```




```
I    32145   C    T    71.32   T|synonymous_varia
I    32511   C    T    110.84  T|synonymous_varia
I    33548   A    C    1064.06 C|missense_variant
I    33612   A    G    572.06  G|synonymous_varia
I    33700   G    A    321.01  A|missense_variant
I    33789   C    G    803.06  G|missense_variant
I    33892   C    G    730.06  G|missense_variant
I    33996   A    G    766.06  G|synonymous_varia
I    34170   T    A    928.06  A|synonymous_varia
I    34308   C    T    452.06  T|synonymous_varia
I    34423   G    A    822.06  A|missense_variant
I    34497   C    T    929.06  T|synonymous_varia
I    34515   T    C    807.06  C|synonymous_varia
I    34554   A    G    973.06  G|synonymous_varia
I    38361   C    T    74.32   T|missense_variant
I    38490   C    G    236.97  G|missense_variant
I    38504   T    G    255.97  G|missense_variant
I    38508   G    T    255.97  T|missense_variant
I    38524   C    A    299.02  A|missense_variant
I    38533   T    C    334.04  C|missense_variant
I    38657   G    A    76.32   A|synonymous_varia
I    38689   G    C    59.32   C|missense_variant
I    38980   G    A    117.84  A|missense_variant
I    39009   T    C    56.84   C|missense_variant
I    40237   C    T    287.02  T|synonymous_varia
I    40380   G    A    104.84  A|synonymous_varia
I    40679   T    C    103.14  C|missense_variant
I    40860   A    G    166.14  G|synonymous_varia
I    40866   T    C    166.14  C|synonymous_varia
I    40914   C    G    196.96  G|missense_variant
I    41278   G    A    111.84  A|missense_variant
I    41481   A    G    235.97  G|synonymous_varia
I    41587   C    T    199.96  T|missense_variant
I    41736   T    C    286.02  C|synonymous_varia
I    41739   A    G    324.04  G|synonymous_varia
I    41752   G    A    239.02  A|missense_variant
I    41802   G    A    243.97  A|synonymous_varia
I    41883   C    T    152.14  T|synonymous_varia
I    41987   C    T    70.32   T|upstream_gene_va
I    42484   G    A    153.14  A|missense_variant
:
```

Step 15 & 16 — Segregate High-Priority Candidates

Filter for variants that disrupt protein synthesis or modify peptide structures:

Bash

```
# Isolate severe loss-of-function variants
grep "HIGH" conA_annotation.txt > conA_HIGH_impact.txt

# Isolate amino-acid substitutions
grep "missense_variant" conA_annotation.txt > conA_missense.txt
```

```
I       64388   C       CT      67.6    CT|frameshift_variant|HIGH|CDC24|YAL041W|transcript|YAL041W_mRNA|protein_coding|1/1|c.1555dupT|p.Ser519fs|1556/2565|
1556/2565|519/854||INFO_REALIGN_3_PRIME,CT|upstream_gene_variant|MODIFIER|PTA1|YAL043C|transcript|YAL043C_mRNA|protein_coding||c.-3337dupA||||3337|,CT|
upstream_gene_variant|MODIFIER|YAL042C-A|YAL042C-A|transcript|YAL042C-A_mRNA|protein_coding||c.-2781dupA||||2781|,CT|downstream_gene_variant|MODIFIER|ERV46|
YAL042W|transcript|YAL042W_mRNA|protein_coding||c.*1825_*1826insT|||||1826|,CT|downstream_gene_variant|MODIFIER|CLN3|YAL040C|transcript|YAL040C_mRNA|
protein_coding||c.*1389dupA|||||1389|,CT|downstream_gene_variant|MODIFIER|CYC3|YAL039C|transcript|YAL039C_mRNA|protein_coding||c.*4327dupA||||4327|
I       144416  C       CA      41.61   CA|frameshift_variant|HIGH|VPS8|YAL002W|transcript|YAL002W_mRNA|protein_coding|1/1|c.718dupA|p.Ile240fs|719/3825|
719/3825|240/1274||INFO_REALIGN_3_PRIME,CA|upstream_gene_variant|MODIFIER|SSA1|YAL005C|transcript|YAL005C_mRNA|protein_coding||c.-2986dupT||||2986|,CA|
downstream_gene_variant|MODIFIER|YAL004W|YAL004W|transcript|YAL004W_mRNA|protein_coding||c.*3009_*3010insA|||||3010|,CA|downstream_gene_variant|MODIFIER|
EFB1|YAL003W|transcript|YAL003W_mRNA|protein_coding||c.*1256_*1257insA||||1257|,CA|downstream_gene_variant|MODIFIER|SNR18|snR18|transcript|snR18_snoRNA|
snoRNA||n.*1948_*1949insA|||||1949|,CA|downstream_gene_variant|MODIFIER|TFC3|YAL001C|transcript|YAL001C_mRNA|protein_coding||c.*3177dupT||||3177|
I       146578  C       CT      50.28   CT|frameshift_variant|HIGH|VPS8|YAL002W|transcript|YAL002W_mRNA|protein_coding|1/1|c.2878dupT|p.Ser960fs|2879/3825|
2879/3825|960/1274||INFO_REALIGN_3_PRIME,CT|downstream_gene_variant|MODIFIER|EFB1|YAL003W|transcript|YAL003W_mRNA|protein_coding||c.*3418_*3419insT|||||
3419|,CT|downstream_gene_variant|MODIFIER|SNR18|snR18|transcript|snR18_snoRNA|snoRNA||n.*4110_*4111insT|||||4111|,CT|downstream_gene_variant|MODIFIER|TFC3|
```

- **Biological Context (Frameshifts):** Indel events in coding regions shift the triplet reading frame, typically resulting in premature termination codons and complete truncation of target proteins (e.g., *CDC24* and *VPS8*).

-
-

```
I	33548	A	C	1334.06	C|missense_variant|MODERATE|BDH2|YAL061W|transcript|YAL061W_mRNA|protein_coding|1/1|c.101A>C|p.Glu34Ala|101/1254|101/1254|34/417||,C|upstream_gene_variant|MODIFIER|BDH1|YAL060W|transcript|YAL060W_mRNA|protein_coding||c.-1607A>C||||1607|,C|upstream_gene_variant|MODIFIER|ECM1|YAL059W|transcript|YAL059W_mRNA|protein_coding||c.-2961A>C||||2961|,C|upstream_gene_variant|MODIFIER|CNE1|YAL058W|transcript|YAL058W_mRNA|protein_coding||c.-3916A>C||||3916|,C|downstream_gene_variant|MODIFIER|GDH3|YAL062W|transcript|YAL062W_mRNA|protein_coding||c.*608A>C||||608|,C|downstream_gene_variant|MODIFIER|YAL059C-A|YAL059C-A|transcript|YAL059C-A_mRNA|protein_coding||c.*2948T>G||||2948|
I	33700	G	A	331.01	A|missense_variant|MODERATE|BDH2|YAL061W|transcript|YAL061W_mRNA|protein_coding|1/1|c.253G>A|p.Gly85Ser|253/1254|253/1254|85/417||,A|upstream_gene_variant|MODIFIER|BDH1|YAL060W|transcript|YAL060W_mRNA|protein_coding||c.-1455G>A||||1455|,A|upstream_gene_variant|MODIFIER|ECM1|YAL059W|transcript|YAL059W_mRNA|protein_coding||c.-2809G>A||||2809|,A|upstream_gene_variant|MODIFIER|CNE1|YAL058W|transcript|YAL058W_mRNA|protein_coding||c.-3764G>A||||3764|,A|downstream_gene_variant|MODIFIER|GDH3|YAL062W|transcript|YAL062W_mRNA|protein_coding||c.*760G>A||||760|,A|downstream_gene_variant|MODIFIER|YAL059C-A|YAL059C-A|transcript|YAL059C-A_mRNA|protein_coding||c.*2796C>T||||2796|,A|downstream_gene_variant|MODIFIER|YAL056C-A|YAL056C-A|transcript|YAL056C-A_mRNA|protein_coding||c.*4996C>T||||4996|
I	33789	C	G	1331.06	G|missense_variant|MODERATE|BDH2|YAL061W|transcript|YAL061W_mRNA|protein_coding|1/1|c.342C>G|p.Asn114Lys|342/1254|342/1254|114/417||,G|upstream_gene_variant|MODIFIER|BDH1|YAL060W|transcript|YAL060W_mRNA|protein_coding||c.-1366C>G||||1366|,G|upstream_gene_variant|MODIFIER|ECM1|YAL059W|transcript|YAL059W_mRNA|protein_coding||c.-2720C>G||||2720|,G|upstream_gene_variant|MODIFIER|CNE1|YAL058W|transcript|YAL058W_mRNA|
```

Biological Context (Missense): Nucleotide transitions alter the single amino acid specified by that codon (such as p.Glu34Ala in *BDH2*), potentially affecting active sites, secondary structures, or protein stability.

2. Heuristic Pileup Calling via VarScan

Step 1 — Prepare the Reference FASTA Index

Generate the coordinate index required by samtools:

Bash

```
samtools faidx GCF_000146045.2_R64_genomic.fa
```

- Produces GCF_000146045.2_R64_genomic.fa.fai for rapid position queries.

Step 2 — Construct the Base Pileup

Compile overlapping reads at every reference base from the sorted BAM file:

Bash

```
samtools mpileup -f GCF_000146045.2_R64_genomic.fa conditionA_Rep2.sorted.bam > aln.raw.pileup
```

- The -f flag supplies the reference sequence to evaluate mismatches against aligned reads in aln.sorted.bam.
- aln.raw.pileup contains positional depth, matching base markers (dots/commas), and base quality ASCII scores.

Step 3 — Discover Raw SNPs

Extract single-nucleotide differences across all pileup sites:

Bash

```
varscan pileup2snp aln.raw.pileup > snp_list
```

```
Warning: No p-value threshold provided, so p-values will not be calculated
Min coverage:   8
Min reads2:     2
Min var freq:   0.01
Min avg qual:   15
P-value thresh: 0.99
Reading input from aln.raw.pileup
7785771 bases in pileup file
1663105 met minimum coverage of 8x
3257 SNPs predicted
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

- Evaluates mismatch ratios against the reference base to flag prospective SNP candidates for subsequent quality filtration.

Step 4 — Discover Raw Indels

Screen for insertions and deletions:

Bash

varscan pileup2indel aln.raw.pileup > indel_list

```
Warning: No p-value threshold provided, so p-values will not be calculated
Min coverage:   8
Min reads2:     2
Min var freq:   0.01
Min avg qual:   15
P-value thresh: 0.99
Reading input from aln.raw.pileup
7785771 bases in pileup file
1663105 met minimum coverage of 8x
530 indels predicted
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

Step 5 — Apply Allele Frequency Filtering

Eliminate low-depth and ambiguous calls by demanding an allele representation of at least 20%:

Bash

varscan pileup2snp aln.raw.pileup --min-var-freq 0.20 > freq_filtered_snps

```
Warning: No p-value threshold provided, so p-values will not be calculated
Min coverage:   8
Min reads2:     2
Min var freq:   0.2
Min avg qual:   15
P-value thresh: 0.99
Reading input from aln.raw.pileup
7785771 bases in pileup file
1663105 met minimum coverage of 8x
2166 SNPs predicted
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

- **Interpretation of Output:** Enforcing --min-var-freq 0.20 discards spurious alignment errors and low-level PCR errors, keeping variants supported by at least one-fifth of covering reads.

Step 6 — Enforce Statistical Significance

Filter calls against Fisher's Exact test to ensure variants are statistically distinct from sequencing noise:

Bash

varscan pileup2snp aln.raw.pileup --p-value 0.1 > pvalue_filtered_snps

```
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$ varscan pileup2snp aln.raw.pileup --p-value 0.1 > pvalue_filtered_snps
Min coverage:   8
Min reads2:     2
Min var freq:   0.01
Min avg qual:   15
P-value thresh: 0.1
Reading input from aln.raw.pileup
7785771 bases in pileup file
1663105 met minimum coverage of 8x
2213 SNPs predicted
(bio-env) (bio-env) ibab@IBAB-MSc-Comp11:~/bio-env$
```

- **Interpretation of Output:** Retained records show p-values well below the 0.1 ceiling , providing mathematical confidence in call authenticity.

3. Microbial Resequencing & Structural Variant Analysis via breseq

Principle & Analytical Framework

breseq is an integrated computational pipeline engineered specifically for bacterial and microbial resequencing. By aligning short sequencing reads against an annotated reference assembly, it evaluates coverage discrepancies, base mismatches, and split-read junctions to predict single nucleotide polymorphisms (SNPs), small indels, structural rearrangements, and mobile-element insertions. The final outputs include machine-readable GenomeDiff (`.gd`) files, standard VCF records, and interactive HTML visual summaries.

Reference Sequence (.fna) + Annotation (.gff) + Sequencing Reads (.fastq.gz)

↓

breseq Engine

↓

Read Alignment & Split-Read Mapping

↓

Coverage & Junction Evidence Analysis

↓

Candidate Mutation Detection

↓

Mutation Assembly & Prediction

↓

GenomeDiff Summary Output (.gd)

↓

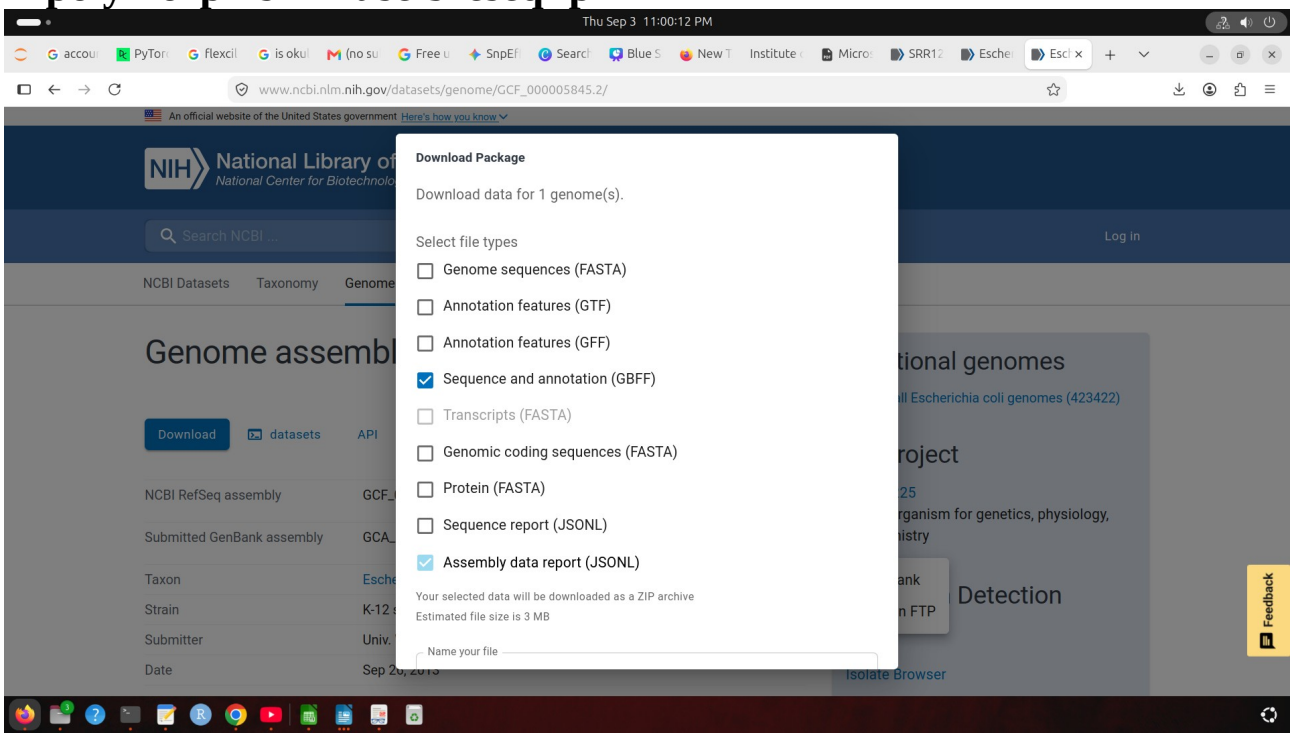
Annotated VCF Output + Visual HTML Report

Breseq----only used for variant call for haploid genomes,like bacteria

If genome contains trasnposons etc,,it can identify that mutations also

Why 40 x depth for clonal and 120x for population?

If polymorphism—use breseq -p

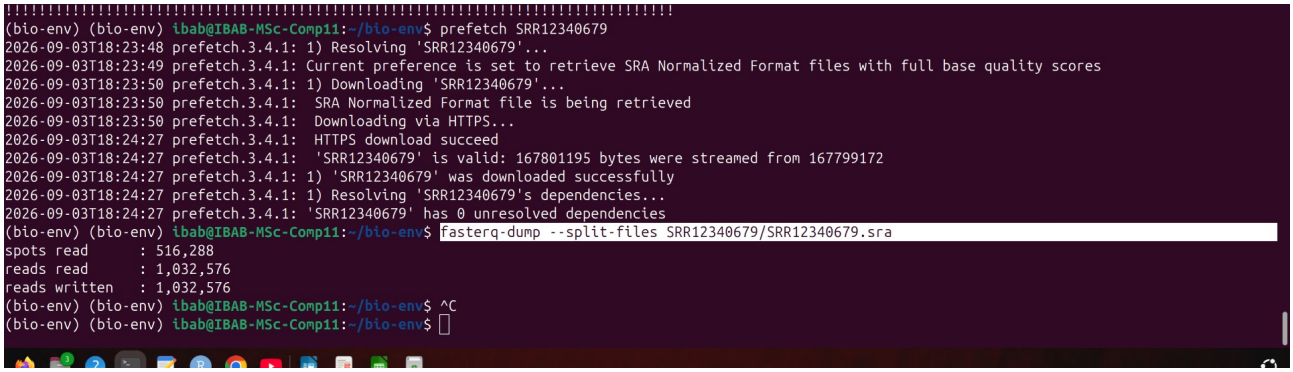


Downloaded SRR12340679 from SRA , Ref U00096 gbff file from NCBI

breseq -r U00096.gbff SRR12340679_1.fastq.gz SRR12340679_2.fastq.gz

prefetch SRR12340679

fasterq-dump --split-files SRR12340679/SRR12340679.sra

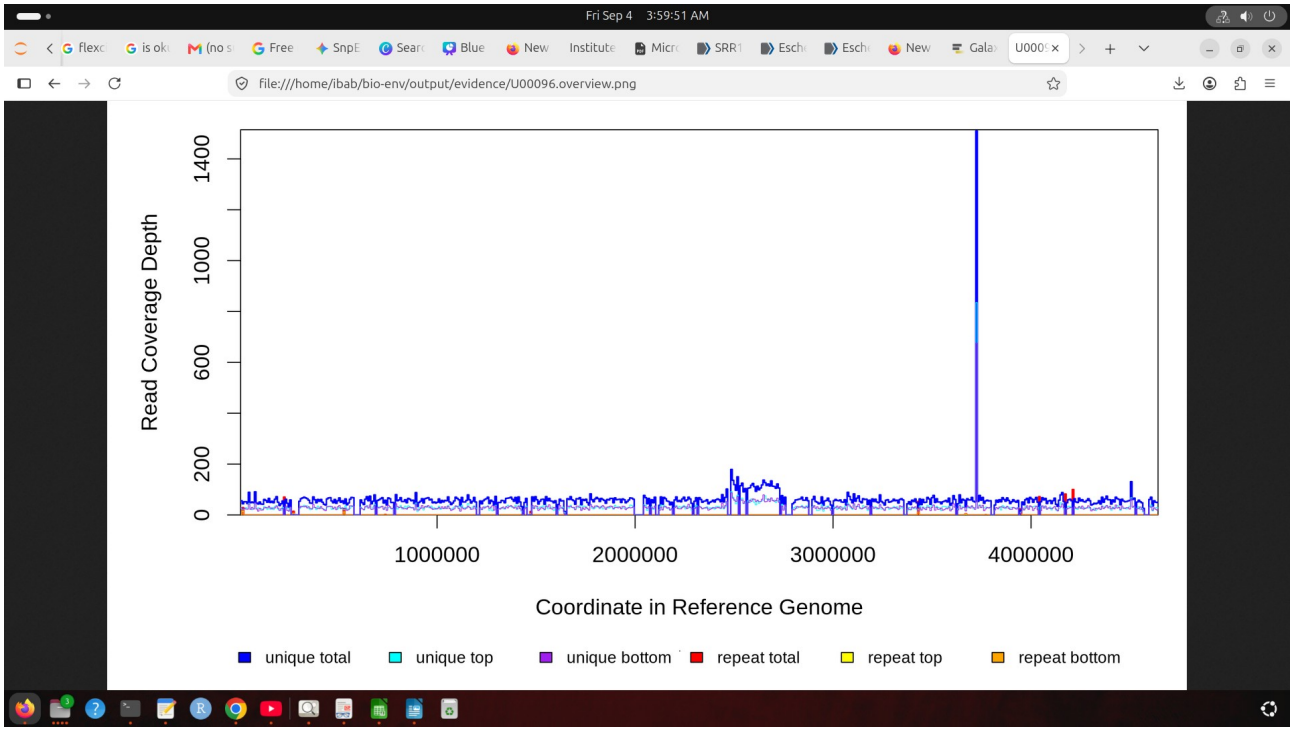


breseq -j 4 -r genomic.gbff SRR12340679_1.fastq SRR12340679_2.fastq

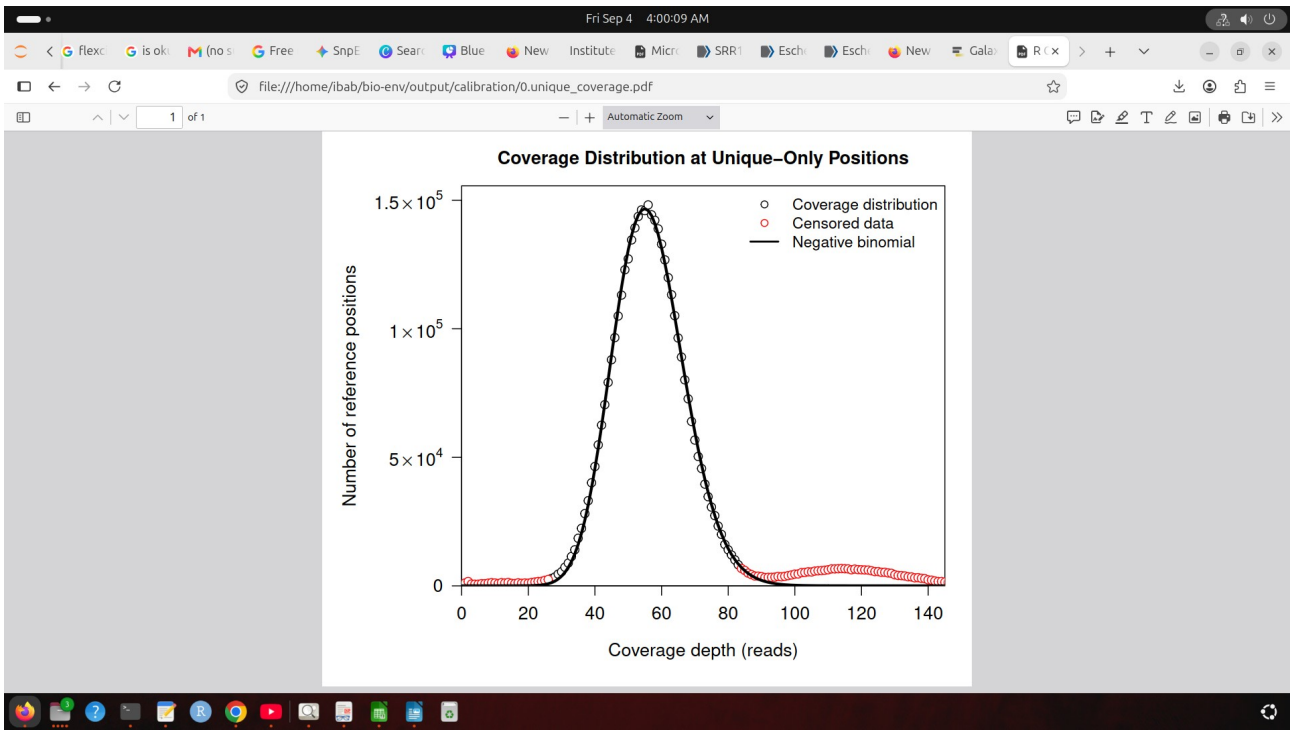
If we see the output we can see why we need genbank file as input because, fasta does not have annotation

output.gd >We can see the htmls...and observe

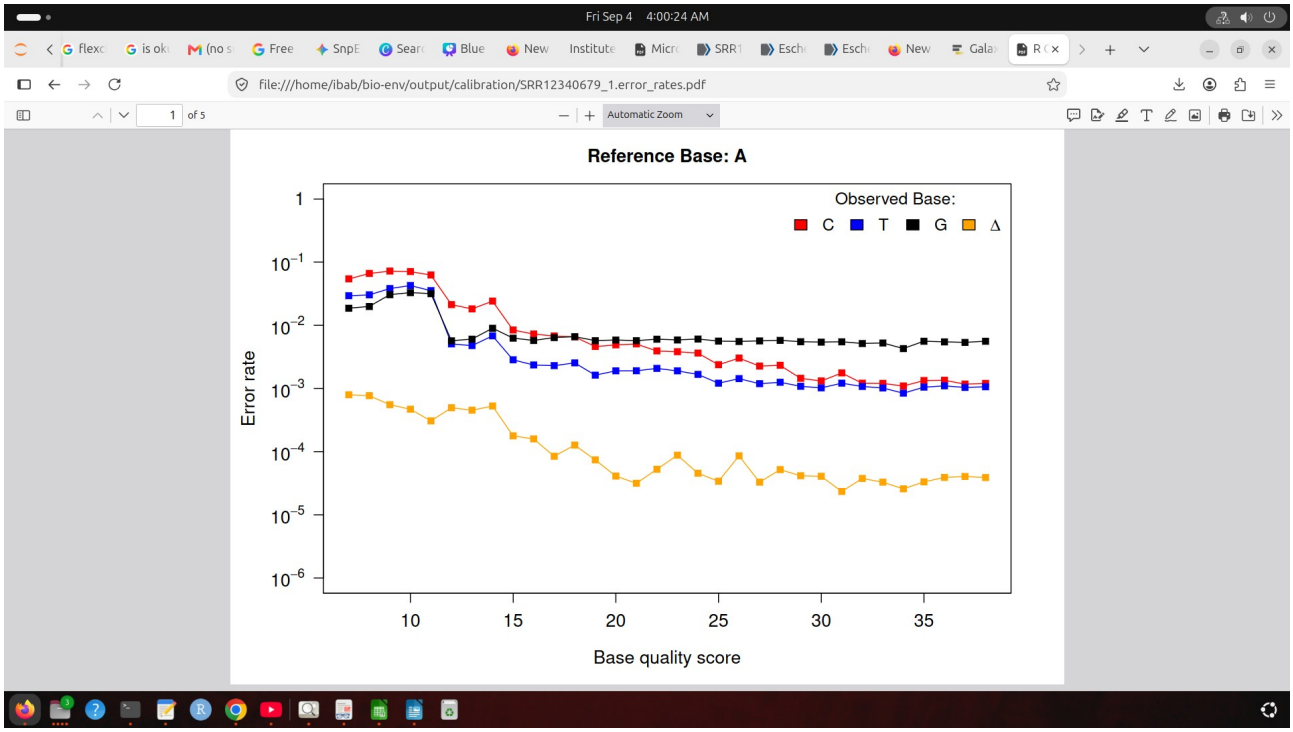
index.htnl



The genome-wide read coverage across the *E. coli* chromosome remains largely uniform at a baseline depth of approximately 50×. A prominent, sharp coverage spike reaching nearly 1500× appears near coordinate 3.75 Mb. This localized surge indicates a high-copy-number genomic duplication, tandem amplification, or repeat-element expansion.

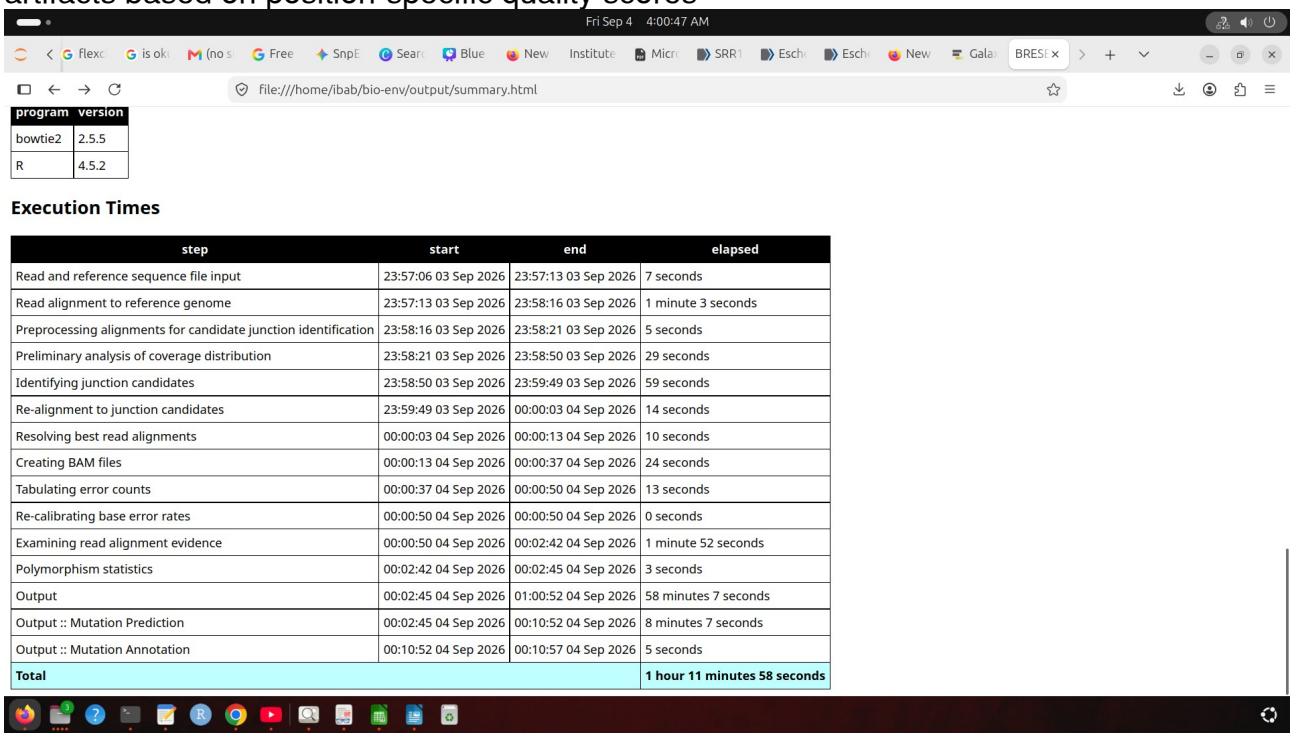


The observed read depth across unique reference positions conforms closely to the fitted Negative Binomial distribution model, centering at a mean coverage of approximately 55 reads. The censored points in the right tail (100–140× coverage) represent duplicated or multi-copy regions that deviate from the single-copy baseline. This model provides a calibrated baseline for breseq to distinguish true deletions and copy-number changes from normal random variation in sequencing depth



The plot displays empirical error rates for reference base **A** across increasing Phred base quality scores. As quality scores improve from Q10 to Q38, observed mismatch frequencies decline steadily, with transitions ($A \rightarrow G$) remaining the most prevalent substitution error, while indels (Δ) remain below 10^{-4}. This

recalibration allows breseq to distinguish real, low-frequency mutations from sequencing artifacts based on position-specific quality scores

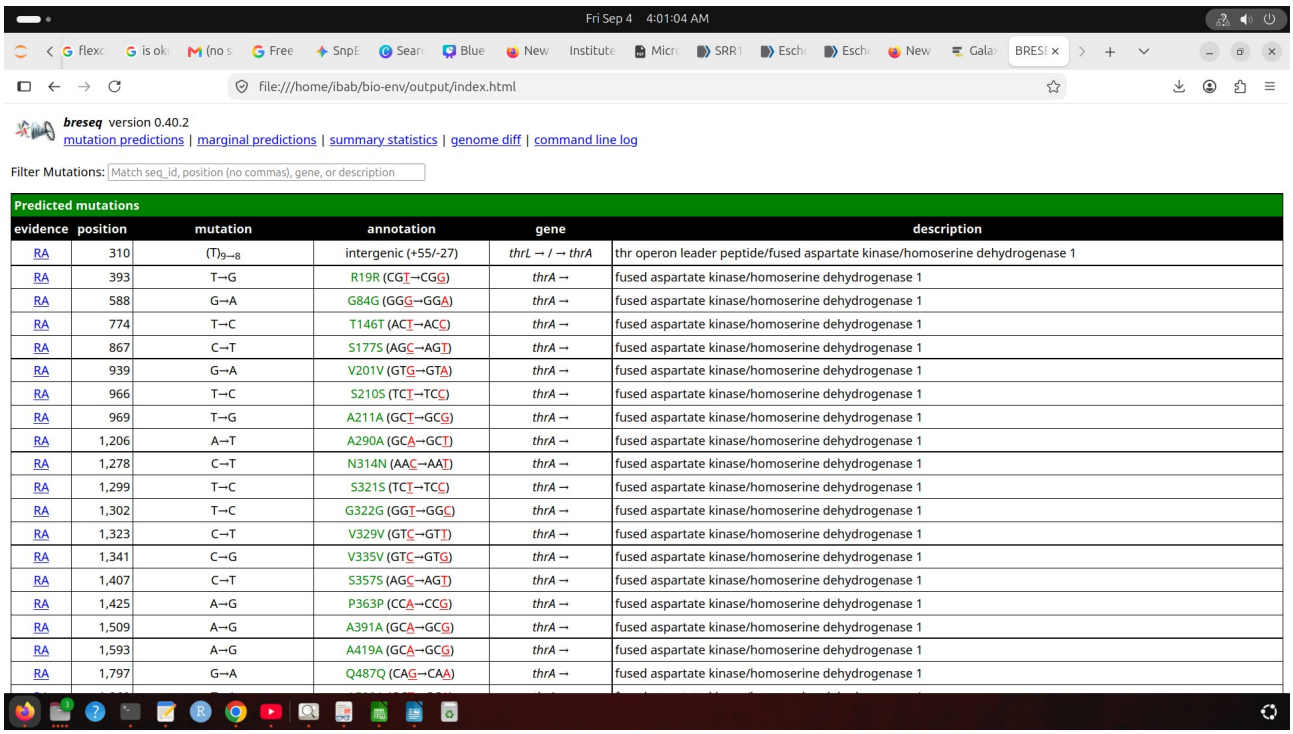


program	version
bowtie2	2.5.5
R	4.5.2

Execution Times

step	start	end	elapsed
Read and reference sequence file input	23:57:06 03 Sep 2026	23:57:13 03 Sep 2026	7 seconds
Read alignment to reference genome	23:57:13 03 Sep 2026	23:58:16 03 Sep 2026	1 minute 3 seconds
Preprocessing alignments for candidate junction identification	23:58:16 03 Sep 2026	23:58:21 03 Sep 2026	5 seconds
Preliminary analysis of coverage distribution	23:58:21 03 Sep 2026	23:58:50 03 Sep 2026	29 seconds
Identifying junction candidates	23:58:50 03 Sep 2026	23:59:49 03 Sep 2026	59 seconds
Re-alignment to junction candidates	23:59:49 03 Sep 2026	00:00:03 04 Sep 2026	14 seconds
Resolving best read alignments	00:00:03 04 Sep 2026	00:00:13 04 Sep 2026	10 seconds
Creating BAM files	00:00:13 04 Sep 2026	00:00:37 04 Sep 2026	24 seconds
Tabulating error counts	00:00:37 04 Sep 2026	00:00:50 04 Sep 2026	13 seconds
Re-calibrating base error rates	00:00:50 04 Sep 2026	00:00:50 04 Sep 2026	0 seconds
Examining read alignment evidence	00:00:50 04 Sep 2026	00:02:42 04 Sep 2026	1 minute 52 seconds
Polymorphism statistics	00:02:42 04 Sep 2026	00:02:45 04 Sep 2026	3 seconds
Output	00:02:45 04 Sep 2026	01:00:52 04 Sep 2026	58 minutes 7 seconds
Output :: Mutation Prediction	00:02:45 04 Sep 2026	00:10:52 04 Sep 2026	8 minutes 7 seconds
Output :: Mutation Annotation	00:10:52 04 Sep 2026	00:10:57 04 Sep 2026	5 seconds
Total			**1 hour 11 minutes 58 seconds**

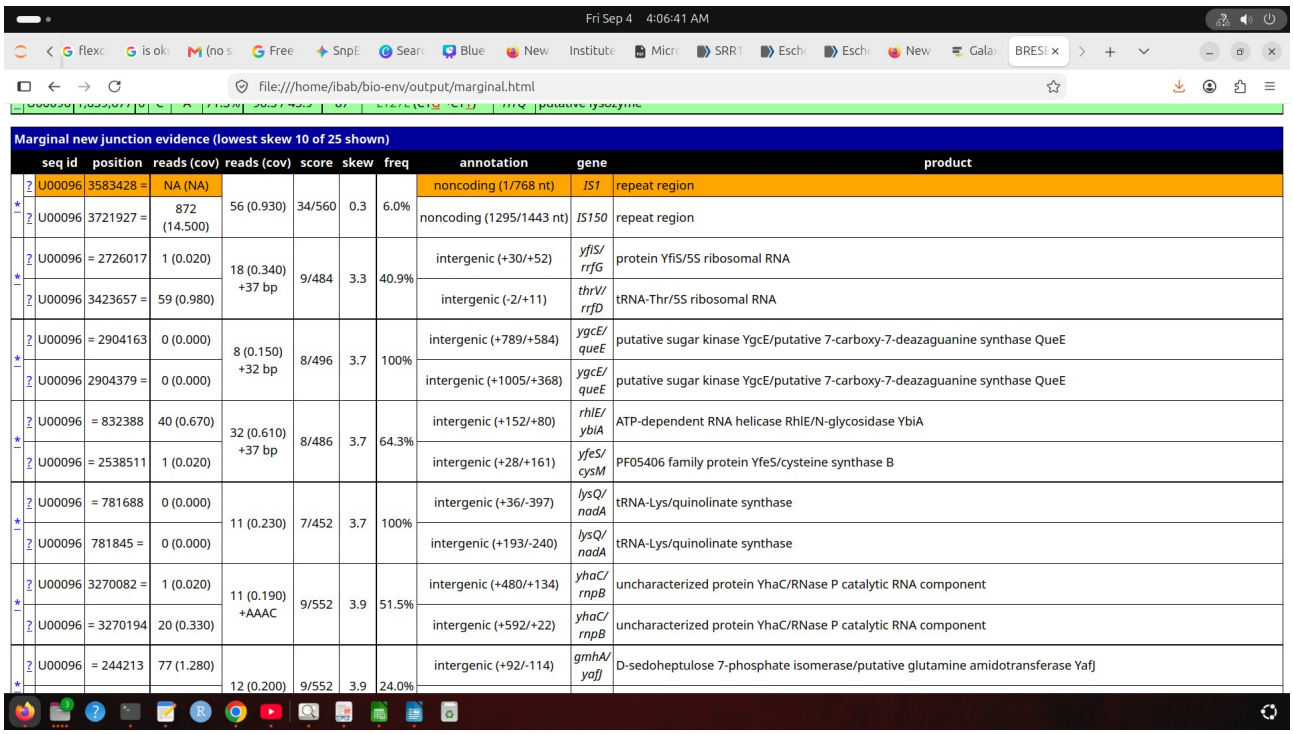
The summary table details the runtime breakdown of the `breseq` run, which finished in **1 hour, 11 minutes, and 58 seconds** using Bowtie2 (v2.5.5) and R (v4.5.2). Read alignment, junction identification, and evidence examination completed rapidly in under 5 minutes combined. The vast majority of processing time was dedicated to final output generation and mutation prediction (~58 minutes), reflecting the computational load of resolving complex genomic evidence across the datase



Predicted mutations

evidence	position	mutation	annotation	gene	description
RA	310	$(T)_{9\to8}$	intergenic (+55/-27)	*thrL* → / → *thrA*	thr operon leader peptide/fused aspartate kinase/homoserine dehydrogenase 1
RA	393	T→G	R19R (CGT→CGG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	588	G→A	G84G (GGG→GGA)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	774	T→C	T146T (ACT→ACC)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	867	C→T	S177S (AGC→AGT)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	939	G→A	V201V (GTG→GTA)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	966	T→C	S210S (TCT→TCC)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	969	T→G	A211A (GCT→GCG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,206	A→T	A290A (GCA→GCT)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,278	C→T	N314N (AAC→AAT)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,299	T→C	S321S (TCT→TCC)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,302	T→C	G322G (GGT→GGC)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,323	C→T	V329V (GTC→GTT)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,341	C→G	V335V (GTC→GTG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,407	C→T	S357S (AGC→AGT)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,425	A→G	P363P (CCA→CCG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,509	A→G	A391A (GCA→GCG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,593	A→G	A419A (GCA→GCG)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1
RA	1,797	G→A	Q487Q (CAG→CAA)	*thrA* →	fused aspartate kinase/homoserine dehydrogenase 1

The table reports high-confidence variants supported by read alignment (RA) evidence, including an intergenic homopolymer contraction $(T)_9 \rightarrow 8$ at position 310. The remaining variants form a dense cluster of single-nucleotide mutations across the *thrA* coding region. Because all *thrA* alterations are synonymous substitutions (e.g., R19R, G84G, T146T), they reflect sequence divergence from the reference without modifying the translated polypeptide.



Marginal new junction evidence (lowest skew 10 of 25 shown)

		seq id	position	reads (cov)	reads (cov)	score	skew	freq	annotation	gene	product
*	?	U00096	3583428 =	NA (NA)	56 (0.930)	34/560	0.3	6.0%	noncoding (1/768 nt)	*IS1*	repeat region
	?	U00096	3721927 =	872 (14.500)					noncoding (1295/1443 nt)	*IS150*	repeat region
*	?	U00096	= 2726017	1 (0.020)	18 (0.340) +37 bp	9/484	3.3	40.9%	intergenic (+30/+52)	*yfiS/ rrfG*	protein YfiS/5S ribosomal RNA
	?	U00096	3423657 =	59 (0.980)					intergenic (-2/+11)	*thrV/ rrfD*	tRNA-Thr/5S ribosomal RNA
*	?	U00096	= 2904163	0 (0.000)	8 (0.150) +32 bp	8/496	3.7	100%	intergenic (+789/+584)	*ygcE/ queE*	putative sugar kinase YgcE/putative 7-carboxy-7-deazaguanine synthase QueE
	?	U00096	2904379 =	0 (0.000)					intergenic (+1005/+368)	*ygcE/ queE*	putative sugar kinase YgcE/putative 7-carboxy-7-deazaguanine synthase QueE
*	?	U00096	= 832388	40 (0.670)	32 (0.610) +37 bp	8/486	3.7	64.3%	intergenic (+152/+80)	*rhlE/ ybiA*	ATP-dependent RNA helicase RhlE/N-glycosidase YbiA
	?	U00096	= 2538511	1 (0.020)					intergenic (+28/+161)	*yfeS/ cysM*	PF05406 family protein YfeS/cysteine synthase B
*	?	U00096	= 781688	0 (0.000)	11 (0.230)	7/452	3.7	100%	intergenic (+36/-397)	*lysQ/ nadA*	tRNA-Lys/quinolinate synthase
	?	U00096	781845 =	0 (0.000)					intergenic (+193/-240)	*lysQ/ nadA*	tRNA-Lys/quinolinate synthase
*	?	U00096	3270082 =	1 (0.020)	11 (0.190) +AAAC	9/552	3.9	51.5%	intergenic (+480/+134)	*yhaC/ rnpB*	uncharacterized protein YhaC/RNase P catalytic RNA component
	?	U00096	= 3270194	20 (0.330)					intergenic (+592/+22)	*yhaC/ rnpB*	uncharacterized protein YhaC/RNase P catalytic RNA component
*	?	U00096	= 244213	77 (1.280)	12 (0.200)	9/552	3.9	24.0%	intergenic (+92/-114)	*gmhA/ yafJ*	D-sedoheptulose 7-phosphate isomerase/putative glutamine amidotransferase YafJ

Junction Skew Score Calculation

reference sequence	pr(no read start)
U00096	0.93658

pr(no read start) is the probability that there will not be an aligned read whose first base matches a given position on a given strand.

Final Junction Predictions

option	value
Coverage evenness (position-hash) score of predicted junctions must be	≥ 3
Minimum probablilty assigned that no mapped read will start at a given position and strand for junction prediction	0.1
Junction allow suboptimal matches	FALSE
Skew score of predicted junction (−log10 probability of unusual coverage evenness) must be	≤ 3
Number of bases that at least one read must overlap each uniquely aligned side of a predicted junction	≥ 1

1. Marginal New Junction Evidence Table

The table identifies candidate breakpoint junctions characteristic of mobile genetic element activity and structural rearrangements, highlighting events in repetitive elements such as **IS1** (position 3,583,428) and **IS150** (position 3,721,927 with 872 supporting reads). Breakpoint calls range from fixed structural junctions (e.g., *ygcE/queE* and *lysQ/nadA*) to subclonal or polymorphic rearrangements (e.g., 24.0% to 64.3%). These split-read alignments resolve genomic reorganizations across intergenic intervals that standard reference-alignment pipelines routinely miss.

2. Junction Skew Score & Filter Criteria

The calculated probability of observing no read starts at a given position and strand is **0.93658** for reference contig U00096, reflecting overall sequencing depth and read start distribution. To differentiate true biological breakpoints from alignment artifacts, breseq applies strict thresholds: a junction coverage, a requirement of at least 1 overlapping base on each uniquely aligned flank. These filters ensure that only junctions with balanced, high-confidence split-read support across both breakpoints are retained as final structural predictions.